

CIS 265 Data Structures and Algorithms

Assignment 1

Due: 12:25 PM Tuesday, September 8, 2026

10 points

In this assignment, you will create a Java program. The program prompts users for numbers between 0 and 100, and calculates the minimum, maximum, and average of the numbers.

I. General Requirements

- 1) The program will keep prompting user to input a valid integer (0-100) until a -1 is input, e.g., "Please input an integer 0-100 (-1 to stop):"
- 2) If an invalid integer is given, the program will print an error message "XXX is not a valid input." where XXX is the given integer, and prompt again.
- 3) After a -1 is input, if no valid number was ever input, i.e., the first input is -1, the program prints "Goodbye!" and exits. Otherwise, the program prints the minimum, maximum, and the average of the integers, and exits.
- 4) You may use any systems or tools to create and run your program. You MUST follow Oracle Java Guide: <https://www.oracle.com/java/technologies/javase/codeconventions-contents.html>
- 5) A sample run will look like this:

```
Please input an integer 0-100 (-1 to stop): 20
Please input an integer 0-100 (-1 to stop): 200
200 is not a valid input.
Please input an integer 0-100 (-1 to stop): 10
Please input an integer 0-100 (-1 to stop): -1
min is 10
max is 20
average is 15
```

II. Submission

This is an individual assignment. Each student needs to submit the source code file of the Java program on Blackboard.

1. Please name your file after your last name. For example, if your last name is "Smith", your file should start with Smith, e.g., SmithAssignment1.java.
2. You need to only submit the source code, i.e., the Java file.
3. You may submit multiple times. Your latest submission before the deadline will be graded.

III. Grading

- 1) A program that does not run will receive 0 points.
- 2) There is a 1-2 point deduction for each major error, e.g., incorrect minimum.
- 3) There is a 1 point deduction for each minor error, e.g., a spelling error in printed message.
- 4) A program that does not follow the guidelines will lose 1 or more points.
- 5) Any type of cheating is not tolerated. You may be asked to explain your program.